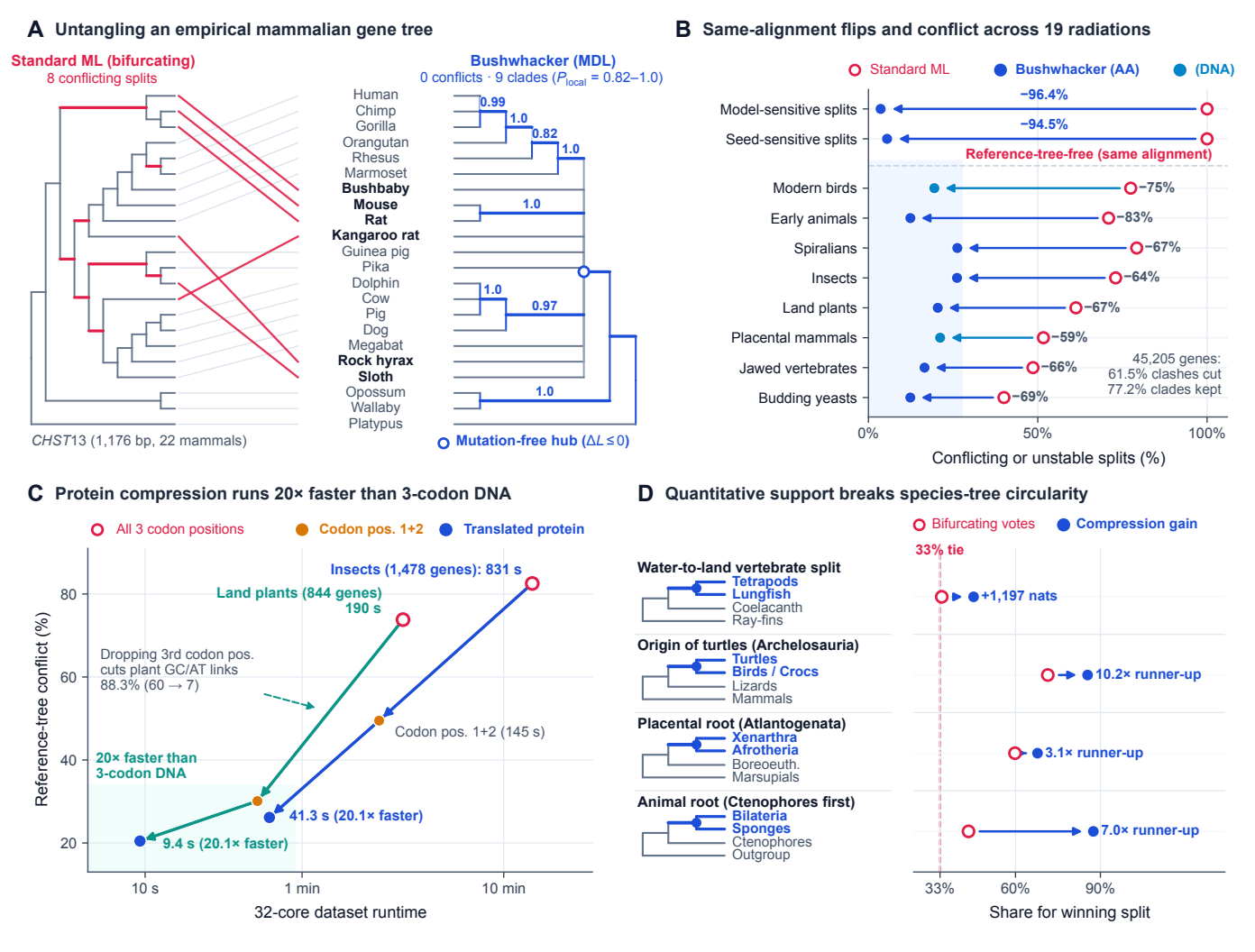

A Untangling an empirical mammalian gene tree
Standard ML (bifurcating)
8 conflicting splits
Bushwhacker (MDL)
0 conflicts · 9 clades (P_local = 0.82–1.0)
Human
Chimp
Gorilla
Orangutan
Rhesus
Marmoset
Bushbaby
Mouse
Rat
Kangaroo rat
Guinea pig
Pika
Dolphin
Cow
Pig
Dog
Megabat
Rock hyrax
Sloth
Opossum
Wallaby
Platypus
0.99
1.0
0.82
1.0
1.0
1.0
0.97
1.0
CHST13 (1,176 bp, 22 mammals)
Mutation-free hub (ΔL ≤ 0)
B Same-alignment flips and conflict across 19 radiations
Standard ML
Bushwhacker (AA)
(DNA)
Model-sensitive splits
Seed-sensitive splits
−96.4%
−94.5%
Reference-tree-free (same alignment)
Modern birds
Early animals
Spiralians
Insects
Land plants
Placental mammals
Jawed vertebrates
Budding yeasts
−75%
−83%
−67%
−64%
−67%
−59%
−66%
−69%
45,205 genes:
61.5% clashes cut
77.2% clades kept
0%
50%
100%
Conflicting or unstable splits (%)
C Protein compression runs 20× faster than 3-codon DNA
All 3 codon positions
Codon pos. 1+2
Translated protein
Insects (1,478 genes): 831 s
Land plants (844 genes)
190 s
Dropping 3rd codon pos.
cuts plant GC/AT links
88.3% (60 → 7)
Codon pos. 1+2 (145 s)
20× faster than
3-codon DNA
41.3 s (20.1× faster)
9.4 s (20.1× faster)
Reference-tree conflict (%)
80
60
40
20
10 s
1 min
10 min
32-core dataset runtime
D Quantitative support breaks species-tree circularity
Bifurcating votes
Compression gain
33% tie
Water-to-land vertebrate split
Tetrapods
Lungfish
Coelacanth
Ray-fins
+1,197 nats
Origin of turtles (Archelosauria)
Turtles
Birds / Crocs
Lizards
Mammals
10.2× runner-up
Placental root (Atlantogenata)
Xenarthra
Afrotheria
Boreoeuth.
Marsupials
3.1× runner-up
Animal root (Ctenophores first)
Bilateria
Sponges
Ctenophores
Outgroup
7.0× runner-up
33%
60%
90%
Share for winning split